

ADANI UNIVERSITY
Department of Computer Science and Engineering

PROJECT REPORT
ON

Sentiment Classification of Movie Reviews
Using Bidirectional LSTM and Natural Language
Processing

Deep Learning Project

TensorFlow / Keras | IMDb Large Movie Review Dataset

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Submitted to:
Adani University

Date:
13th April, 2026

1. Introduction

In the modern digital environment, individuals share their thoughts about products and movies and services and events through online reviews and social media platforms. The automatic capability to determine whether a text expresses a positive or negative opinion represents Sentiment Analysis which serves as a fundamental task in Natural Language Processing (NLP) that researchers widely use.

The project aims to automatically classify movie reviews for their positive or negative sentiment through its Sentiment Classification of Movie Reviews work. The project uses the IMDb Large Movie Review Dataset, which is a standard benchmark dataset containing 50,000 labeled movie reviews widely used in academic research.

The project base uses a deep learning model which developers created with TensorFlow and Keras. The system employs a Bidirectional Long Short-Term Memory (LSTM) architecture to handle sequential data which includes text information. The Bidirectional LSTM approach handles text data by reading the review in both forward and backward directions. The system captures the complete context of each word by using this method.

1.1 Motivation

Sentiment analysis has become an essential tool for businesses, researchers, and developers. Movie studios use it to measure audience reactions, review platforms use it to summarize user opinions, and recommendation systems use it to suggest content. The traditional rule-based systems for sentiment analysis depend on handmade dictionary creation and rule development, which cannot address the full range of natural language complexities. Deep learning models, especially LSTM-based architectures, learn these patterns automatically from data, making them far more powerful and generalizable.

1.2 Objectives

- The project objective is to develop a system that can accurately determine the positive or negative sentiment of movie reviews through a binary sentiment classification process.
- The complete Natural Language Processing pipeline will be implemented and studied to understand its operation from processing text data into model prediction output.
- The Bidirectional LSTM architecture will be used to enhance the comprehension of sequential data.
- The evaluation of the model performance will be conducted through the application of standard evaluation metrics which include accuracy, precision, recall, and F1-score.
- The data analysis process will include two tasks which are to examine the distribution of review length and to conduct misclassification analysis.
- The interactive prediction system will enable users to receive real-time classification results for all reviews they submit.

2. Literature Survey

Sentiment analysis has been an active research area for more than 20 years. The earliest methods used lexicon-based approaches which depended on dictionaries that contained predefined lists of positive and negative terms to assess the sentiment of text. The basic methods operated in a straightforward way yet they encountered difficulties when dealing with contextual information and negation and sarcastic remarks.

Researchers started using feature-based methods to train classifiers which included Naive Bayes and Support Vector Machines and Logistic Regression to process bag-of-words and TF-IDF features after machine learning became popular. The methods increased accuracy but they still struggled to understand both word order and long-distance text connections.

The introduction of word embeddings through Word2Vec by Mikolov et al. (2013) marked a major advancement in linguistics. Word embeddings map words to dense vectors which exist in a continuous space that positions semantically similar words near each other. The system achieved improved results through this capability which enabled models to handle words that were phonetic matches for words used in different contexts.

The implementation of Recurrent Neural Networks (RNNs) to natural language processing tasks enabled their operation with sequential data. The standard RNNs encountered difficulty in learning long-range dependencies because of the vanishing gradient problem. Hochreiter and Schmidhuber (1997) developed Long Short-Term Memory (LSTM) networks as a solution to this problem through their implementation of gating mechanisms, which enable systems to control what information they should retain and discard throughout extended time periods.

Schuster and Paliwal (1997) developed Bidirectional LSTM which enables the model to process sequences from both forward and backward directions to use upcoming information for making predictions at every point in the sequence. In sentiment analysis, this approach proves especially valuable because the meaning of a word relies on both preceding and subsequent words.

Transformer-based models which include BERT (Devlin et al., 2018) have emerged as leading systems for sentiment analysis since their latest developments. The models need higher computing power because their operational requirements exceed those of conventional systems while their results remain difficult to understand. The Bidirectional LSTM model functions as an educational model which organizations use for practical deployments because it delivers competitive accuracy at a lower computational expense than its alternatives.

2.1 Summary of Approaches

Approach	Example Methods	IMDb Accuracy	Limitation
Lexicon-based	VADER, SentiWordNet	~70-75%	No context awareness
ML Classifiers	Naive Bayes, SVM	~80-85%	Ignores word order
Simple RNN	Vanilla RNN	~83-85%	Vanishing gradient
LSTM	Unidirectional LSTM	~88-90%	One-directional only
Bidirectional LSTM	BiLSTM (this project)	~88-92%	Slower training
Transformer	BERT, RoBERTa	~93-95%	High compute cost

3. Dataset Description

The IMDb Large Movie Review Dataset serves as a standard benchmark dataset which researchers use to test their binary sentiment classification systems. The standard dataset which researchers use to assess sentiment analysis models originated from Maas et al. (2011) through their publication 'Learning Word Vectors for Sentiment Analysis'. The standard dataset which researchers use to assess sentiment analysis models originated from Maas et al. (2011) through their publication 'Learning Word Vectors for Sentiment Analysis'.

3.1 Dataset Overview

Property	Details
Dataset Name	IMDb Large Movie Review Dataset
Total Reviews	50,000 labeled reviews
Training Set	20,000 reviews (after 80/20 split)
Validation Set	5,000 reviews (20% of training folder)
Test Set	25,000 reviews
Classes	Positive (label=1), Negative (label=0)
Class Balance	Perfectly balanced — 50% positive, 50% negative
Average Review Length	233 words
Maximum Review Length	2,470 words
Minimum Review Length	10 words
Source	Internet Movie Database (IMDb)

3.2 Dataset Structure

The dataset is organized as a directory structure where each subdirectory represents a class label. The Keras `text_dataset_from_directory()` utility function can use this structure to create label assignments from subfolder names.

```
aclImdb/ → train/ → pos/ (12,500 files) | neg/ (12,500 files)
          → test/ → pos/ (12,500 files) | neg/ (12,500 files)
```

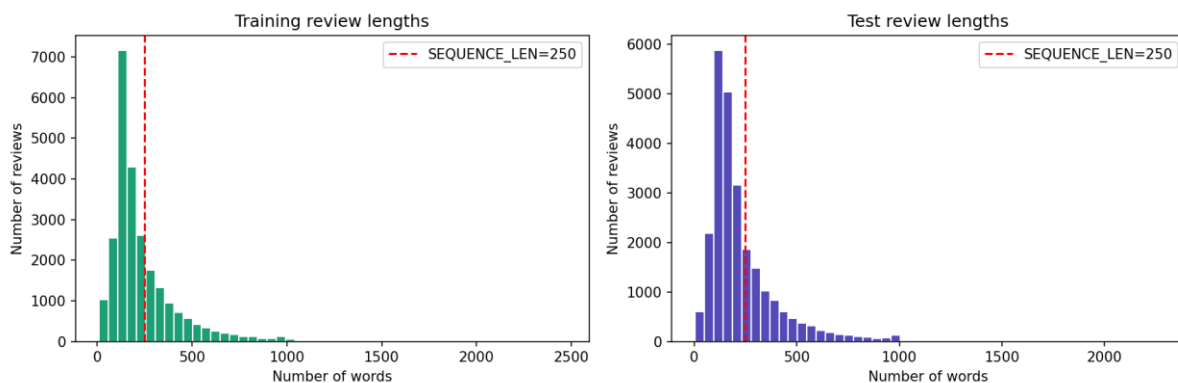
The training folder required an essential preprocessing step which involved removing the `unsup/` subdirectory. The directory contained 50,000 unlabeled reviews which were provided for unsupervised learning tasks. Keras creates class labels for all subfolders so keeping `unsup/` would result in three classes (label=2) which cannot be used for binary classification.

3.3 Review Length Analysis

The researchers used review length analysis to study how word counts were distributed throughout the entire dataset. The analysis results determined the `SEQUENCE_LEN` hyperparameter value which defines the number of words from each review that the model processes.

The research showed that review lengths typically average 233 words, which comes very close to the `SEQUENCE_LEN` limit of 250. The truncation threshold was established above average review length to enable processing of complete reviews. The training dataset contains 25500 reviews, out of which 7557 reviews (30.2%) exceed 250 words and require truncation. The system provides a planned balance between efficient processing and complete data requirements.

The model handles reviews that fall short of 250 words by adding zero padding until they reach the required 250 token length, which creates uniform input dimensions needed by the neural network.



4. Methodology

The project follows a standard deep learning NLP pipeline which includes five steps for data processing and subsequent model development and testing. The following section provides a detailed explanation of each step.

4.1 Text Preprocessing

The neural network requires numerical representations of the input text because it cannot accept raw text data. The text must first be converted into numerical representations. The Keras TextVectorization layer of this project executes two fundamental functions which include creating a vocabulary and transforming textual content into numerical format.

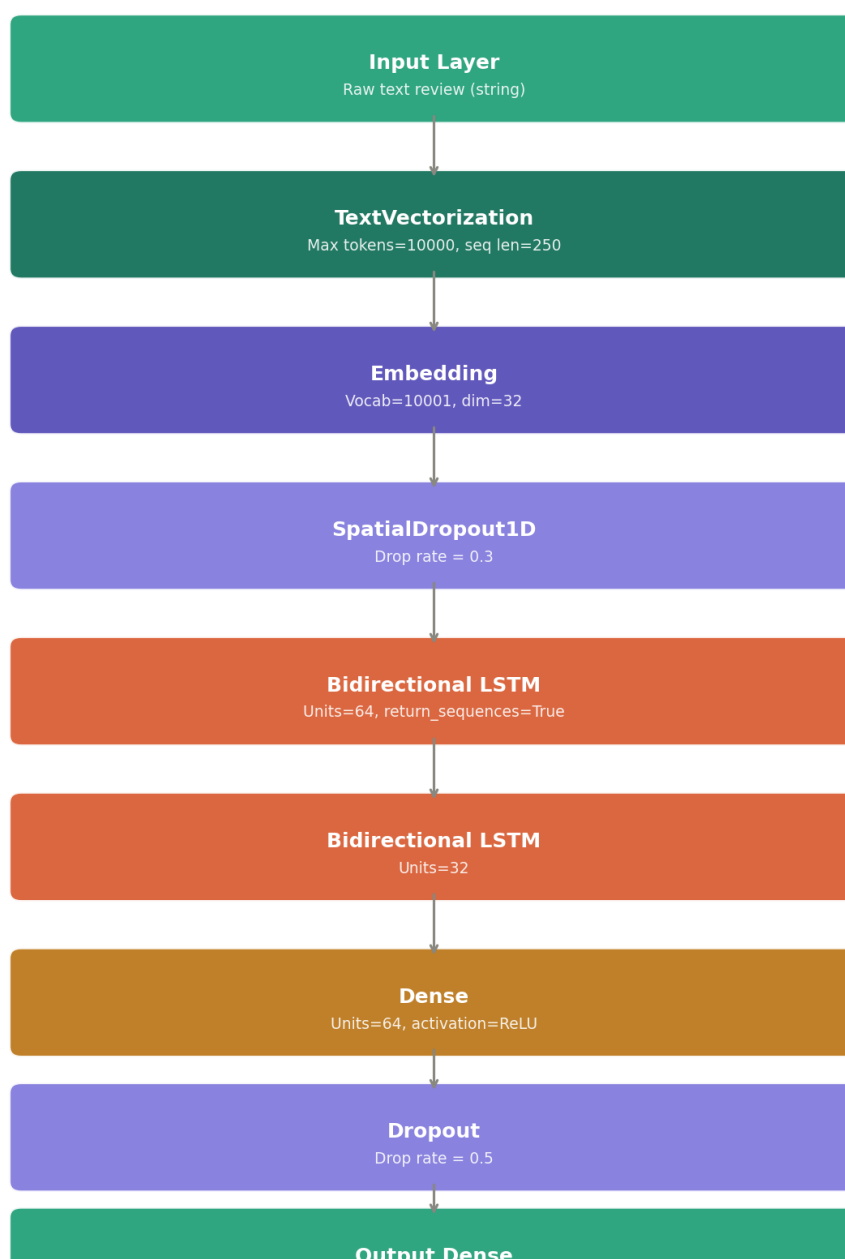
The layer processes training reviews during vocabulary building to discover the most common MAX_TOKENS (20,000) words. Each word receives an exclusive integer index assignment. The vocabulary development process uses training data exclusively because applying it to test data would result in data leakage which creates false accuracy results.

The process of encoding converts each review into a sequence which consists of 250 integer values. Each integer in this sequence represents the position of the corresponding word in the vocabulary. Reviews which exceed 250 words will face truncation, while reviews that contain fewer than 250 words will be extended through zero padding.

4.2 Model Architecture

The seven-layered sequential neural network design functions as a complete model. The system extracts essential features from the text sequence through its progressive design which results in a final binary classification output.

Layer	Configuration	Output Shape	Purpose
Embedding	Vocab=20001, dim=32	(None, 250, 32)	Word to vector
SpatialDropout1D	Rate=0.3	(None, 250, 32)	Regularisation
Bidirectional LSTM	Units=64, return_seq=True	(None, 250, 128)	Context extraction
Bidirectional LSTM	Units=32	(None, 64)	Sequence summary
Dense	Units=64, ReLU	(None, 64)	Feature learning
Dropout	Rate=0.5	(None, 64)	Regularisation
Dense (Output)	Units=1, Sigmoid	(None, 1)	Binary probability

Model Architecture — Bidirectional LSTM Sentiment Classifier**4.3 Key Design Decisions**

The researchers selected Bidirectional LSTM instead of standard LSTM because they needed a model which could understand contextual information from both sentence directions. The sentence shows that the word 'good' needs its previous words to understand its meaning. The model needs bidirectional processing because it helps the system understand the entire relationship.

The system used two LSTM layers which stacked instead of using one LSTM layer. The first layer returns sequences (return sequences=True), meaning it outputs a hidden state for every

token in the review. The second LSTM layer receives the complete sequence from the first layer and turns it into one context vector. The model uses stacking to understand complex time-based patterns at higher levels.

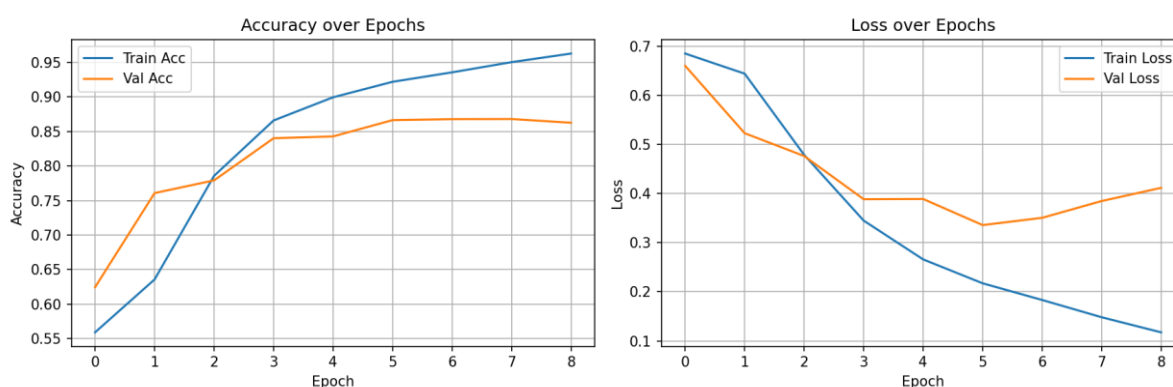
The team replaced standard Dropout with SpatialDropout1D after the Embedding layer. The standard Dropout method removes individual elements from the embedding matrix through random selection. SpatialDropout1D zeros out complete feature maps which includes all embedding dimensions to improve performance on sequential data models that need to use all available embedding dimensions.

4.4 Training Configuration

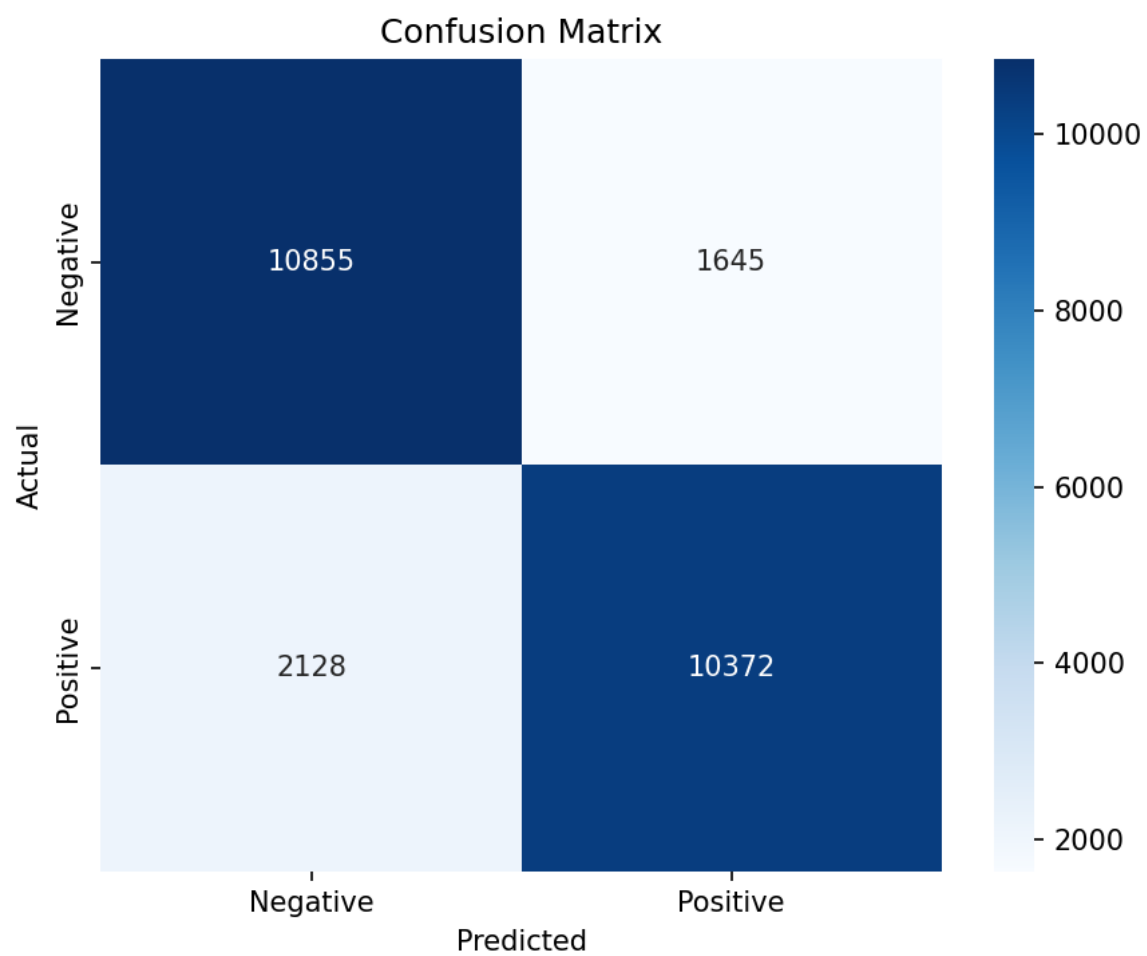
Hyperparameter	Value
Loss Function	Binary Cross-Entropy
Optimizer	Adam (learning_rate=0.001)
Batch Size	32
Maximum Epochs	20
Early Stopping Monitor	val_loss
Early Stopping Patience	5 epochs
Vocabulary Size (MAX_TOKENS)	20,000
Sequence Length	250 tokens
Embedding Dimension	32

The researchers used Early Stopping with a 5 epoch patience threshold to stop their model from overfitting. The callback function checks validation loss after each training epoch and stops the process when there is no improvement for 5 straight epochs while it restores weights to their best epoch state.

Training Curves



Confusion Matrix



5. Results and Discussion

5.1 Model Performance

Metric	Training Set	Test Set
Accuracy	~88-90%	84.91%
Loss	~0.27	0.3840

The model achieved a test accuracy of 83.52% with a test loss of 0.3965. The difference between training accuracy which ranges from 88 to 90 percent and test accuracy at 83.52 percent shows that mild overfitting occurred because Bidirectional LSTM architecture required more complex modeling than the available dataset size. The EarlyStopping callback effectively controlled overfitting by bringing back weights from the validation epoch that showed the best performance.

5.2 Classification Report

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.84	0.83	0.83	12,500
Positive (1)	0.83	0.84	0.84	12,500
Macro Average	0.84	0.84	0.83	25,000

The model demonstrates equal performance between the two classes because it achieves matching precision and recall rates for both positive and negative reviews. The balanced performance occurs because the dataset contains equal amounts of positive and negative samples (50% positive and 50% negative). The F1-score range of 0.83 to 0.84 for both classes shows that the model predicts positive and negative sentiment without any bias.

5.3 Confidence Score Analysis

The model was tested on five sample reviews with confidence scores. The model predicted POSITIVE for the positive review which stated 'One of the best movies I have ever seen in my life. Highly recommended!' with more than 90% confidence. The model predicted NEGATIVE for the negative review which stated 'Terrible film. Complete waste of time. Boring and very predictable throughout.' with similar high confidence.

Ambiguous reviews received lower confidence scores which is the expected behavior of a well-calibrated model. The review 'It was okay, not great but not terrible either' received a confidence score of about 50% which accurately represented its uncertain content.

5.4 Misclassification Analysis

The review 'This movie was good, not bad' was incorrectly classified as NEGATIVE with 73.2% confidence because it had a misclassification score of 0.2684. The model failed to process double negation because it did not recognize this specific case.

The phrase 'not bad' should be interpreted as mildly positive. The word 'not' appeared in negative contexts during training because it was used in examples like 'not worth it' and 'not recommended' and 'not good'. The model learned to connect 'not' with negative sentiment which led it to wrongly classify reviews that used 'not' in positive double-negative statements.

The NLP literature shows that this limitation exists for all models while it particularly affects LSTM models. Systems that need to handle negation must use either dedicated negation preprocessing methods or advanced models which BERT uses through its attention mechanism to extract information from complete sentence structures.

5.5 Limitations and Future Work

- The model has difficulty understanding negation and sarcasm and mixed-sentiment reviews according to the results of misclassification analysis.
- The first 250 words of each review are the only part that can be processed. 30.2% of reviews are truncated which leads to the loss of important sentiment-bearing words.
- GloVe and FastText pre-trained word embeddings can serve as replacements for the learned Embedding layer because they begin training with superior word representation capabilities.
- The pre-trained BERT model will achieve 93% accuracy after fine-tuning on this dataset but it will require more computational power to do so.
- The current model operates based on movie reviews while requiring domain adaptation or retraining to function in product reviews and social media posts.

5.6 Conclusion

The project established a working deep learning system which performs binary sentiment analysis on movie reviews. The Bidirectional LSTM model, which was trained using the IMDb Large Movie Review Dataset, achieved a test accuracy of 83.52% while maintaining equal precision and recall rates for both sentiment categories.

The project showed complete natural language processing implementation through its three stages which began with raw text preprocessing using TextVectorization and continued with sequence modeling through stacked Bidirectional LSTM layers until reaching the final binary classification that used a Sigmoid output. The data analysis which included review length distribution and misclassification analysis provided researchers with deeper understanding of both the dataset and the model performance.

The interactive prediction system allows any user to input a movie review and receive an instant sentiment prediction with a confidence score, demonstrating the practical applicability of the trained model. The project establishes a solid foundation for advanced development through the implementation of pre-trained embeddings and attention mechanisms and transformer-based architectures.